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Strut assembly and method for coupling an engine to a wing of an aircraft

Abstract

A strut assembly for coupling an engine to a wing of an aircraft, the strut assembly comprising a mid truss assembly comprising a fore portion and an aft portion; an engine mount member connected to the fore portion of the mid truss assembly; an aft truss assembly comprising a fore portion and an aft portion, the fore portion of the aft truss assembly being connected to the aft portion of the mid truss assembly; an aft strut bulkhead connected to the aft portion of the aft truss assembly; and a wing mounting feature operatively connected to, and located aft of, the aft strut bulkhead.

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Background/Summary

PRIORITY (1) This application claims priority from U.S. Ser. No. 63/364,636 filed on May 13, 2022.

FIELD

(1) The present disclosure is generally related to and, more specifically, to strut assemblies and associated methods for coupling an engine to a wing of an aircraft.

BACKGROUND

(2) Typical structures for mounting engines to wings of aircraft incorporate a design wherein the strut box sits on top of the engine. This strut box typically attaches to the engine in two locations, generally on the engine fan case and on the turbine rear frame. Because the strut box is on top of the engine, the engine generally has to be shifted downward to accommodate structural height clearance. Further, engine bending is problematic and commonly observed in that it drives stiffness and weight into engines because the thrust is reacted off of the engine centerline.

(3) Accordingly, those skilled in the art continue research and development in the field of coupling engines to wings of aircraft and associated methods for coupling engines of wings to aircraft.

SUMMARY

(4) Disclosed are strut assemblies for coupling an engine to a wing of an aircraft.

(5) In an example, the strut assembly includes a mid truss assembly comprising a fore portion and an aft portion, an engine mount member connected to the fore portion of the mid truss assembly, an aft truss assembly comprising a fore portion and an aft portion, the fore portion of the aft truss assembly being connected to the aft portion of the mid truss assembly, an aft strut bulkhead connected to the aft portion of the aft truss assembly, and a wing mounting feature operatively connected to, and located aft of, the aft strut bulkhead.

(6) Also disclosed is an aircraft.

(7) In an example, the aircraft includes a wing, an engine located below the wing, a strut assembly coupling the engine to the wing, the strut assembly includes a mid truss assembly, an aft truss assembly connected to the mid truss assembly, an aft strut bulkhead connected to the aft portion of the aft truss assembly, wing mounting features located aft of the aft strut bulkhead, and an engine mount member connected to a fore portion of the mid truss assembly.

(8) Also disclosed is a method for connecting an engine to a wing of an aircraft.

(9) In an example, the method includes positioning a strut assembly between the engine and the wing. The strut assembly includes a fore truss assembly, a mid truss assembly connected to the fore truss assembly, an aft truss assembly connected to the mid truss assembly, an aft strut bulkhead connected to the aft portion of the aft truss assembly, wing mounting features located aft of the aft strut bulkhead, and an engine mount member connected to the fore truss assembly.

(10) Other examples of the disclosed strut assemblies, aircraft, and methods will become apparent from the following detailed description, the accompanying drawings, and the appended claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a side schematic of a strut assembly coupling an engine to a wing of an aircraft;
- (2) FIG. 2 is a front schematic of the strut assembly of FIG. 1;
- (3) FIG. 3 is a front schematic of the strut assembly of FIG. 1;
- (4) FIG. 4 is a perspective schematic of the strut assembly of FIG. 1;
- (5) FIG. 5 is a top view schematic of the strut assembly of FIG. 1;
- (6) FIG. 6 is a side schematic of a strut assembly coupling an engine to a wing of an aircraft;
- (7) FIG. 7 is a front schematic of the strut assembly of FIG. 6;
- (8) FIG. 8 is a perspective schematic of the strut assembly of FIG. 6;
- (9) FIG. 9 is a top view schematic of the strut assembly of FIG. 6;
- (10) FIG. 10 is a flow chart of a method for coupling an engine to a wing of an aircraft;
- (11) FIG. 11 is a flow diagram of an aircraft manufacturing and service methodology; and
- (12) FIG. 12 is a schematic block diagram of an example of an aircraft.

DETAILED DESCRIPTION

(13) The following detailed description refers to the accompanying drawings, which illustrate specific examples described by the present disclosure. Other examples having different structures and operations do not depart from the scope of the present disclosure. Like reference numerals may refer to the same feature, element, or component in the different drawings.

(14) Illustrative, non-exhaustive examples, which may be, but are not necessarily, claimed, of the subject matter according to the present disclosure are provided below. Reference herein to “example” means that one or more feature, structure, element, component, characteristic, and/or operational step described in connection with the example is included in at least one aspect, embodiment, and/or implementation of the subject matter according to the present disclosure. Thus, the phrases “an example,” “another example,” “one or more examples,” and similar language throughout the present disclosure may, but do not necessarily, refer to the same example. Further, the subject matter characterizing any one example may, but does not necessarily, include the subject matter characterizing any other example. Moreover, the subject matter characterizing any one example may be, but is not necessarily, combined with the subject matter characterizing any other example.

(15) Referring to FIG. 1-FIG. 5, disclosed is a strut assembly **130** for coupling an engine **120** to a wing **110** of an aircraft **100**, FIG. 12. The strut assembly **130** may include a portion of a pylon **175**. The strut assembly **130** may be configured to mount an unducted fan engine or an open rotor engine to a wing **110** of an aircraft **100**. The strut assembly **130** may be configured to mount to one or two station planes on an engine **120**, based upon mount vibration affects. Using the disclosed strut assembly **130**, thrust can be reacted at or below an engine centerline, alleviating engine bending, and further allowing for lighter engines. The disclosed strut assembly **130** design may further allow movement up to 17.5” up as compared to convention strut to wing architectures.

(16) In one example, the strut assembly **130** includes a mid truss assembly **140** having a fore portion **144** and an aft portion **146**, see FIG. 1. In one or more examples, the mid truss assembly **140** may include a plurality of mid truss members **142**. In one example, each mid truss member **142'** of the plurality of mid truss members **142** is hollow. In another example, each mid truss member **142'** of the plurality of mid truss members **142** is corrosion resistant. The plurality of mid truss members **142** may be of any material have requisite material properties for the strut assembly **130**. In one example, each mid truss member **142'** of the plurality of mid truss members **142** comprises a metallic material. In another example, each mid truss member **142'** of the plurality of mid truss members **142** comprises steel. In another example, each mid truss member **142'** of the plurality of mid truss members **142** comprises a titanium alloy. In another example, each mid truss member **142'** of the plurality of mid truss members **142** comprises Ti-6Al-4V. In yet another

example, each mid truss member **142'** of the plurality of mid truss members **142** comprises Inconel. In yet a further example, each mid truss member **142'** of the plurality of mid truss members **142** comprises Ti-6Al-2Sn-4Zr-2Mo.

(17) Referring to FIG. **1**, in one or more examples, the strut assembly **130** further includes an aft truss assembly **150** having a fore portion **156** and an aft portion **154**. The fore portion **156** of the aft truss assembly **150** is connected to the aft portion **146** of the mid truss assembly **140**. In one example, the aft truss assembly **150** comprises a plurality of aft truss members **152**. In one example, each aft truss member **152'** of the plurality of aft truss members **152** is hollow. In another example, each aft truss member **152'** of the plurality of aft truss members **152** is corrosion resistant. The plurality of aft truss members **152** may be of any material have requisite material properties for the strut assembly **130**. In one example, each aft truss member **152'** of the plurality of aft truss members **152** comprises a metallic material. In another example, each aft truss member **152'** of the plurality of aft truss members **152** comprises steel. In another example, each aft truss member **152'** of the plurality of aft truss members **152** comprises a titanium alloy. In another example, each aft truss member **152'** of the plurality of aft truss members **152** comprises Ti-6Al-4V. In yet another example, each aft truss member **152'** of the plurality of aft truss members **152** comprises Inconel. In yet a further example, each aft truss member **152'** of the plurality of aft truss members **152** comprises Ti-6Al-2Sn-4Zr-2Mo.

(18) Referring to FIG. **4**, in one or more examples, the strut assembly **130** further includes an aft strut bulkhead **160** connected to the aft portion **154** of the aft truss assembly **150**. In one example, the aft strut bulkhead **160** comprises a fireproof material and serves as an engine compartment firewall.

(19) Referring to FIG. **4**, in one or more examples, the strut assembly **130** further includes a wing mounting **164** feature operatively connected to, and located aft of, the aft strut bulkhead **160**. The wing mounting feature **164** may be configured to couple with an outside surface of the wing **110**. In one example, the wing mounting feature **164** may be configured to couple with a top surface of the wing **110**.

(20) The strut assembly **130** further includes an engine mount member **180** connected to the fore portion **144** of the mid truss assembly **140**. The engine mount member **180** may be of any shape and configuration suitable for the aircraft **100**, FIG. **12**. In one example, the engine mount member **180** comprises a ring shape configured to fully encircle a portion of the engine **120** when the strut assembly **130** is connected to the engine **120**. In another example, the engine mount member **180** comprises a half-ring shape configured to partially encircle a portion of the engine **120** when the strut assembly **130** is connected to the engine **120**. In yet another example, the engine mount member **180** comprises a plurality of engine mounting features **182**. Each engine mounting feature **182'** of the plurality of engine mounting features **182** may be configured to dampen vibration.

(21) Referring to FIG. **3** and FIG. **5**, in one or more examples, the strut assembly **130** further includes a mid strut bulkhead **170** between the aft truss assembly **150** and the mid truss assembly **140**.

(22) Referring to FIG. **6**-FIG. **9**, in one or more examples, the strut assembly **130** further includes a fore truss assembly **190** connected to the engine mount member **180**. In one example, the fore truss assembly **190** includes a plurality of fore truss members **192**. Still referring to FIG. **6**, in one or more examples, the strut assembly **130** further comprising a fore engine mounting feature **194** connected to the fore truss assembly **190**.

(23) Referring to FIG. **12**, disclosed is an aircraft **100**. In one or more examples, the aircraft **100** includes a wing **110**, an engine **120** located below the wing **110m** and a strut assembly **130** coupling the engine **120** to the wing **110**.

(24) Referring to FIG. **1**, the strut assembly **130** of the aircraft **100** includes a mid truss assembly **140** having a fore portion **144** and an aft portion **146**. In one or more examples, the mid truss assembly **140** may include a plurality of mid truss members **142**. In one example, each mid truss

member **142'** of the plurality of mid truss members **142** is hollow. In another example, each mid truss member **142'** of the plurality of mid truss members **142** is corrosion resistant. The plurality of mid truss members **142** may be of any material have requisite material properties for the strut assembly **130**. In one example, each mid truss member **142'** of the plurality of mid truss members **142** comprises a metallic material. In another example, each mid truss member **142'** of the plurality of mid truss members **142** comprises steel. In another example, each mid truss member **142'** of the plurality of mid truss members **142** comprises a titanium alloy. In another example, each mid truss member **142'** of the plurality of mid truss members **142** comprises Ti-6Al-4V. In yet another example, each mid truss member **142'** of the plurality of mid truss members **142** comprises Inconel. In yet a further example, each mid truss member **142'** of the plurality of mid truss members **142** comprises Ti-6Al-2Sn-4Zr-2Mo.

(25) Referring to FIG. **1**, the strut assembly **130** of the aircraft **100** includes an aft truss assembly **150** connected to the mid truss assembly **140**. The aft truss assembly **150** may have a fore portion **156** and an aft portion **154**. The fore portion **156** of the aft truss assembly **150** is connected to the aft portion **146** of the mid truss assembly **140**. In one example, the aft truss assembly **150** comprises a plurality of aft truss members **152**. In one example, each aft truss member **152'** of the plurality of aft truss members **152** is hollow. In another example, each aft truss member **152'** of the plurality of aft truss members **152** is corrosion resistant. The plurality of aft truss members **152** may be of any material have requisite material properties for the strut assembly **130**. In one example, each aft truss member **152'** of the plurality of aft truss members **152** comprises a metallic material. In another example, each aft truss member **152'** of the plurality of aft truss members **152** comprises steel. In another example, each aft truss member **152'** of the plurality of aft truss members **152** comprises a titanium alloy. In another example, each aft truss member **152'** of the plurality of aft truss members **152** comprises Ti-6Al-4V. In yet another example, each aft truss member **152'** of the plurality of aft truss members **152** comprises Inconel. In yet a further example, each aft truss member **152'** of the plurality of aft truss members **152** comprises Ti-6Al-2Sn-4Zr-2Mo.

(26) Referring to FIG. **4**, the strut assembly **130** of the aircraft **100** includes an aft strut bulkhead **160** connected to the aft portion **154** of the aft truss assembly **150**. In one example, the aft strut bulkhead **160** comprises a fireproof material. The aft strut bulkhead **160** may further serve as an engine compartment firewall.

(27) Referring to FIG. **4**, the strut assembly **130** of the aircraft **100** includes wing mounting features **164** located aft of the aft strut bulkhead **160**. The wing mounting feature **164** may be configured to couple with an outside surface of the wing **110**. In one example, the wing mounting feature **164** may be configured to couple with a top surface of the wing **110**.

(28) Referring to FIG. **2**, in one or more examples, the strut assembly **130** of the aircraft **100** further includes an engine mount member **180** connected to a fore portion **144** of the mid truss assembly **140**. The engine mount member **180** may be of any shape and configuration suitable for the aircraft **100**. In one example, the engine mount member **180** comprises a ring shape configured to fully encircle a portion of the engine **120** when the strut assembly **130** is connected to the engine **120**. In another example, the engine mount member **180** comprises a half-ring shape configured to partially encircle a portion of the engine **120** when the strut assembly **130** is connected to the engine **120**. In yet another example, the engine mount member **180** comprises a plurality of engine mounting features **182**. Each engine mounting feature **182'** of the plurality of engine mounting features **182** may be configured to dampen vibration.

(29) Referring to FIG. **3**, in one or more examples, the strut assembly **130** of the aircraft **100** further includes a mid strut bulkhead **170** located between the aft truss assembly **150** and the mid truss assembly **140**. The strut assembly **130** of the aircraft **100** may further include a pylon **175** located between the wing **110** and the engine **120** of the aircraft **100**. In one example, the strut assembly **130** defines a portion of the pylon **175**.

(30) Referring to FIG. 6-FIG. 9, in one or more examples, the strut assembly **130** of the aircraft **100** further includes a fore truss assembly **190** connected to the engine mount member **180**. In one example, the fore truss assembly **190** comprises a plurality of fore truss members **192**. The strut assembly **130** of the aircraft **100** may further include a fore engine mounting feature **194** connected to the fore truss assembly **190**.

(31) Referring to FIG. 10, disclosed is a method **200** for connecting an engine **120** to a wing **110** of an aircraft **100**. In one or more examples, the method **200** includes positioning **210** a strut assembly **130** between the engine **120** and the wing **110**. In one example, the strut assembly **130** includes a fore truss assembly **190**. The fore truss assembly **190** may include a plurality of fore truss members **192**. The strut assembly **130** of the method **200** may further include a fore engine mounting feature **194** connected to the fore truss assembly **190**.

(32) Referring to FIG. 1, in one or more examples, the strut assembly **130** of the method **200** includes a mid truss assembly **140** connected to the fore truss assembly **190**. The mid truss assembly **140** may include a plurality of mid truss members **142**. In one example, each mid truss member **142'** of the plurality of mid truss members **142** is hollow. In another example, each mid truss member **142'** of the plurality of mid truss members **142** is corrosion resistant. The plurality of mid truss members **142** may be of any material have requisite material properties for the strut assembly **130**. In one example, each mid truss member **142'** of the plurality of mid truss members **142** comprises a metallic material. In another example, each mid truss member **142'** of the plurality of mid truss members **142** comprises steel. In another example, each mid truss member **142'** of the plurality of mid truss members **142** comprises a titanium alloy. In another example, each mid truss member **142'** of the plurality of mid truss members **142** comprises Ti-6Al-4V. In yet another example, each mid truss member **142'** of the plurality of mid truss members **142** comprises Inconel. In yet a further example, each mid truss member **142'** of the plurality of mid truss members **142** comprises Ti-6Al-2Sn-4Zr-2Mo.

(33) Referring to FIG. 1, in one or more examples, the strut assembly **130** of the method **200** includes an aft truss assembly **150** connected to the mid truss assembly **140**. The aft truss assembly **150** may include a plurality of aft truss members **152**. In one example, each aft truss member **152'** of the plurality of aft truss members **152** is hollow. In another example, each aft truss member **152'** of the plurality of aft truss members **152** is corrosion resistant. The plurality of aft truss members **152** may be of any material have requisite material properties for the strut assembly **130**. In one example, each aft truss member **152'** of the plurality of aft truss members **152** comprises a metallic material. In another example, each aft truss member **152'** of the plurality of aft truss members **152** comprises steel. In another example, each aft truss member **152'** of the plurality of aft truss members **152** comprises a titanium alloy. In another example, each aft truss member **152'** of the plurality of aft truss members **152** comprises Ti-6Al-4V. In yet another example, each aft truss member **152'** of the plurality of aft truss members **152** comprises Inconel. In yet a further example, each aft truss member **152'** of the plurality of aft truss members **152** comprises Ti-6Al-2Sn-4Zr-2Mo.

(34) Referring to FIG. 4, in one or more examples, the strut assembly **130** of the method **200** includes an aft strut bulkhead **160** connected to the aft portion **154** of the aft truss assembly **150**. In one example, the aft strut bulkhead **160** is fireproof.

(35) Referring to FIG. 4, in one or more examples, the strut assembly **130** of the method **200** includes wing mounting features **164** located aft of the aft strut bulkhead **160**. and an engine mount member **180** connected to the fore truss assembly **190**. The wing mounting feature **164** may be configured to couple with an outside surface of the wing **110**. In one example, the wing mounting feature **164** may be configured to couple with a top surface of the wing **110**.

(36) Referring to FIG. 10, in one or more examples, the method **200** further includes mounting **220** the strut assembly **130** to the engine **120** and mounting **230** the strut assembly **130** to the wing **110** of the aircraft **100**. The mounting **220** may include coupling the engine mount member **180** to the

engine **120** with the plurality of engine mounting features **182**. The engine mount member **180** may be of any shape and configuration suitable for the aircraft **100**. In one example, the engine mount member **180** comprises a ring shape configured to fully encircle a portion of the engine **120** when the strut assembly **130** is connected to the engine **120**. In another example, the engine mount member **180** comprises a half-ring shape configured to partially encircle a portion of the engine **120** when the strut assembly **130** is connected to the engine **120**. In yet another example, the engine mount member **180** comprises a plurality of engine mounting features **182**. Each engine mounting feature **182'** of the plurality of engine mounting features **182** may be configured to dampen vibration.

(37) Referring to FIG. 5, in one or more example, the strut assembly **130** of the method **200** further includes a mid strut bulkhead **170** located between the aft truss assembly **150** and the mid truss assembly **140**.

(38) Other benefits of the disclosed strut assembly **130** include reduction in back pressure to the fan, allowance for lower total wetted area (total surface area exposed) for engine **120**, pylon **175** and wing **110**, reduction in engine **120** bending due to thrust, pitch and yaw, engine **120** fire compartment firewall simplification, increased volume available for systems routing and systems integration, architecture versatility, allowance for multiple engine mounting architectures, engine mounting architecture can be tailored to enhance engine aspects such as: SFC, spatial integration, engine vibration isolation for cabin environment, allowance for optimum engine placement for either a high or low wing aircraft **100**, and locating the engine centerline closer to the wing **110**, allowing for less landing gear growth for fan blade tip clearance to ground.

(39) In one aspect, the strut assembly **130** has a gradual increase in cross sectional area, which aids in reduction of back-pressure variation on the engine fan. This design further improves engine fan efficiency and reduces fuel burn, vibration, and noise. The disclosed strut assembly **130** further allows for a fairing shape that has lower total wetted area (total surface area exposed) for the aircraft, providing reduced drag. The disclosed strut assembly **130** may have an open truss-type cradle design without using a typical strut box design that allows the space to be used for system routing. Further benefits of the disclosed design include aerodynamics as the proposed strut assembly **130** allows for improved aerodynamics based upon engine configuration. Additionally, the strut assembly **130** includes an aft bulkhead that acts as a firewall to prevent fires from propagating to the rest of the aircraft, simplifying fire safety methodology and further isolating vibration within the aircraft.

(40) Examples of the present disclosure may be described in the context of aircraft manufacturing and service method **1100** as shown in FIG. 11 and aircraft **1102** as shown in FIG. 12. During pre-production, service method **1100** may include specification and design (block **1104**) of aircraft **1102** and material procurement (Block **1106**). During production, component and subassembly manufacturing (Block **1108**) and system integration (Block **1110**) of aircraft **1102** may take place. Thereafter, aircraft **1102** may go through certification and delivery (Block **1112**) to be placed in service (Block **1114**). While in service, aircraft **1102** may be scheduled for routine maintenance and service (Block **1116**). Routine maintenance and service may include modification, reconfiguration, refurbishment, etc. of one or more systems of aircraft **1102**.

(41) Each of the processes of service method **1100** may be performed or carried out by a system integrator, a third party, and/or an operator (e.g., a customer). For the purposes of this description, a system integrator may include, without limitation, any number of aircraft manufacturers and major-system subcontractors; a third party may include, without limitation, any number of vendors, subcontractors, and suppliers; and an operator may be an airline, leasing company, military entity, service organization, and so on.

(42) As shown in FIG. 12, aircraft **1102** produced by service method **1100** may include airframe **1118** with a plurality of high-level systems **1120** and interior **1122**. Examples of high-level systems **1120** include one or more of propulsion system **1124**, electrical system **1126**, hydraulic system

1128, and environmental system **1130**. Any number of other systems may be included. Although an aerospace example is shown, the principles disclosed herein may be applied to other industries, such as the automotive industry. Accordingly, in addition to aircraft **1102**, the principles disclosed herein may apply to other vehicles, e.g., land vehicles, marine vehicles, space vehicles, etc.

(43) Strut assembly (ies), aircraft, and method(s) shown or described herein may be employed during any one or more of the stages of the manufacturing and service method **1100**. For example, components or subassemblies corresponding to component and subassembly manufacturing (block **1108**) may be fabricated or manufactured in a manner similar to components or subassemblies produced while aircraft **1102** is in service (Block **1114**). Also, one or more examples of the strut assembly (ies), aircraft, and method(s), or combination thereof may be utilized during production stages component and subassembly manufacturing (Block **1108**) and system integration (Block **1110**), for example, by substantially expediting assembly of or reducing the cost of aircraft **1102**. Similarly, one or more examples of the system or method realizations, or a combination thereof, may be utilized, for example and without limitation, while aircraft **1102** is in service (Block **1114**) and/or during maintenance and service (Block **1116**).

(44) Different examples of the strut assembly (ies), aircraft, and method(s) disclosed herein include a variety of components, features, and functionalities. It should be understood that the various examples of the strut assembly (ies), aircraft, and method(s) disclosed herein may include any of the components, features, and functionalities of any of the other examples of the strut assembly (ies), aircraft, and method(s) disclosed herein in any combination, and all of such possibilities are intended to be within the scope of the present disclosure.

(45) Many modifications of examples set forth herein will come to mind to one skilled in the art to which the present disclosure pertains having the benefit of the teachings presented in the foregoing descriptions and the associated drawings.

(46) Therefore, it is to be understood that the present disclosure is not to be limited to the specific examples illustrated and that modifications and other examples are intended to be included within the scope of the appended claims. Moreover, although the foregoing description and the associated drawings describe examples of the present disclosure in the context of certain illustrative combinations of elements and/or functions, it should be appreciated that different combinations of elements and/or functions may be provided by alternative implementations without departing from the scope of the appended claims. Accordingly, parenthetical reference numerals in the appended claims are presented for illustrative purposes only and are not intended to limit the scope of the claimed subject matter to the specific examples provided in the present disclosure.

Claims

1. A strut assembly for coupling an engine to a wing of an aircraft, the strut assembly comprising: a mid strut bulkhead; an engine mount member spaced away from the mid strut bulkhead; an aft strut bulkhead spaced away from the mid strut bulkhead, opposite the engine mount member; a mid truss assembly comprising: a central mid truss member extending between the mid strut bulkhead and the engine mount member; an inboard mid truss member extending between the mid strut bulkhead and the engine mount member; and an outboard mid truss member extending between the mid strut bulkhead and the engine mount member; wherein the central mid truss member, the inboard mid truss member, and the outboard mid truss member diverge in a forward direction from the mid strut bulkhead to the engine mount member; an aft truss assembly extending between the mid strut bulkhead and the aft strut bulkhead; a wing mounting feature operatively connected to, and located aft of, the aft strut bulkhead; a fore truss assembly consisting essentially of: a central fore truss member positioned at an upper center of the engine mount member and extending forward from the engine mount member and aligned with the central mid truss member; an inboard fore truss member angularly spaced away from the central fore truss member along the engine

mount member and extending forward from the engine mount member; and an outboard fore truss member angularly spaced away from the central fore truss member along the engine mount member, opposite the inboard fore truss member, and extending forward from the engine mount member, wherein the central fore truss member, the inboard fore truss member, and the outboard fore truss member converge in a forward direction from the engine mount member to a forward node, opposite the engine mount member, that forms a terminus of a fore portion of the fore truss assembly; an inboard engine mounting feature coupled directly to the engine mount member and angularly spaced away from the inboard fore truss member, opposite the central fore truss member, along the engine mount member; an outboard engine mounting feature coupled directly to the engine mount member and angularly spaced away from the outboard fore truss member, opposite the central fore truss member, along the engine mount member; and a fore engine mounting feature coupled directly to the fore truss assembly, positioned at the forward node of the fore truss assembly, and aligned with the central fore truss member.

2. The strut assembly of claim 1, wherein the engine mount member comprises a half-ring shape configured to partially encircle a portion of the engine when the strut assembly is connected to the engine.

3. The strut assembly of claim 1, wherein the aft strut bulkhead comprises a fireproof material.

4. The strut assembly of claim 1, wherein the mid truss assembly further comprises additional mid truss members extending between the mid strut bulkhead and the engine mount member.

5. The strut assembly of claim 4, wherein each mid truss member of the mid truss assembly is hollow.

6. The strut assembly of claim 4, wherein each mid truss member of the mid truss assembly is corrosion resistant.

7. The strut assembly of claim 4, wherein each mid truss member of the mid truss assembly comprises a metallic material.

8. The strut assembly of claim 4, wherein each mid truss member of the mid truss assembly comprises one or more of a steel material, a titanium alloy, Ti-6Al-4V, and Ti-6Al-2Sn-4Zr-2Mo.

9. The strut assembly of claim 1, wherein the aft truss assembly comprises a plurality of aft truss members.

10. The strut assembly of claim 9, wherein each aft truss member of the aft truss assembly is hollow.

11. The strut assembly of claim 9, wherein each aft truss member of the aft truss assembly is corrosion resistant.

12. The strut assembly of claim 9, wherein each aft truss member of the aft truss assembly comprises a metallic material.

13. The strut assembly of claim 1, wherein the inboard engine mounting feature, the outboard engine mounting feature, and the fore engine mounting feature are configured to dampen vibrations.

14. An aircraft comprising: a wing; an engine located below the wing; and a strut assembly coupling the engine to the wing, the strut assembly comprising: a mid strut bulkhead; an engine mount member spaced away from the mid strut bulkhead; an aft strut bulkhead spaced away from the mid strut bulkhead, opposite the engine mount member; a mid truss assembly comprising: a central mid truss member extending between and connected to the mid strut bulkhead and the engine mount member; an inboard mid truss member extending between and connected to the mid strut bulkhead and the engine mount member; and an outboard mid truss member extending between and connected to the mid strut bulkhead and the engine mount member; wherein the central mid truss member, the inboard mid truss member, and the outboard mid truss member diverge in a forward direction from the mid strut bulkhead to the engine mount member; an aft truss assembly extending between and connected to the mid strut bulkhead and the aft strut bulkhead; wing mounting features connected to and located aft of the aft strut bulkhead; a fore truss

assembly consisting essentially of: a central fore truss member positioned at an upper center of the engine mount member and extending forward from the engine mount member and aligned with the central mid truss member; an inboard fore truss member angularly spaced away from the central fore truss member along the engine mount member and extending forward from the engine mount member; and an outboard fore truss member angularly spaced away from the central fore truss member along the engine mount member, opposite the inboard fore truss member, and extending forward from the engine mount member, wherein the central fore truss member, the inboard fore truss member, and the outboard fore truss member converge in a forward direction from the engine mount to a forward node, opposite the engine mount member, that forms a terminus of a fore portion of the fore truss assembly; an inboard engine mounting feature coupled directly to the engine mount member and angularly spaced away from the inboard fore truss member, opposite the central fore truss member, along the engine mount member; an outboard engine mounting feature coupled directly to the engine mount member and angularly spaced away from the outboard fore truss member, opposite the central fore truss member, along the engine mount member; and a fore engine mounting feature coupled directly to the fore truss assembly, positioned at the forward node of the fore truss assembly, and aligned with the central fore truss member.

15. The aircraft of claim 14, wherein the inboard engine mounting feature, the outboard engine mounting feature, and the fore engine mounting feature are configured to dampen vibrations from the engine.

16. The aircraft of claim 14, wherein the engine mount member comprises a half-ring shape configured to partially encircle a portion of the engine when the strut assembly is connected to the engine.

17. A method for connecting an engine to a wing of an aircraft, the method comprising: positioning a strut assembly between the engine and the wing, the strut assembly comprising: a mid strut bulkhead; an engine mount member spaced away from the mid strut bulkhead; an aft strut bulkhead spaced away from the mid strut bulkhead, opposite the engine mount member; a mid truss assembly comprising: a central mid truss member extending between the mid strut bulkhead and the engine mount member; an inboard mid truss member extending between the mid strut bulkhead and the engine mount member; and an outboard mid truss member extending between the mid strut bulkhead and the engine mount member; wherein the central mid truss member, the inboard mid truss member, and the outboard mid truss member diverge in a forward direction from the mid strut bulkhead to the engine mount member; an aft truss assembly connected to and extending between the mid strut bulkhead and the aft strut bulkhead; wing mounting features connected to and located aft of the aft strut bulkhead; and a fore truss assembly consisting essentially of: a central fore truss member positioned at an upper center of the engine mount member and extending forward from the engine mount member and aligned with the central mid truss member; an inboard fore truss member angularly spaced away from the central fore truss member along the engine mount member and extending forward from the engine mount member; and an outboard fore truss member angularly spaced away from the central fore truss member along the engine mount member, opposite the inboard fore truss member, and extending forward from the engine mount member, wherein the central fore truss member, the inboard fore truss member, and the outboard fore truss member converge in a forward direction from the engine mount to a forward node, opposite the engine mount member, that forms a terminus of a fore portion of the fore truss assembly; an inboard engine mounting feature coupled directly to the engine mount member and angularly spaced away from the inboard fore truss member, opposite the central fore truss member, along the engine mount member; an outboard engine mounting feature coupled directly to the engine mount member and angularly spaced away from the outboard fore truss member, opposite the central fore truss member, along the engine mount member; and a fore engine mounting feature coupled to the fore truss assembly, positioned at the forward node of the fore truss assembly, and aligned with the central fore truss member; coupling the strut assembly to the wing; and coupling

the strut assembly to the engine at the inboard engine mounting feature, the outboard engine mounting feature, and the fore engine mounting feature.

18. The method of claim 17, further comprising damping vibrations from the engine using the inboard engine mounting feature, the outboard engine mounting feature, and the fore engine mounting feature.

19. The strut assembly of claim 14, wherein each mid truss member of the mid truss assembly and each aft truss member of the aft truss assembly is hollow.
